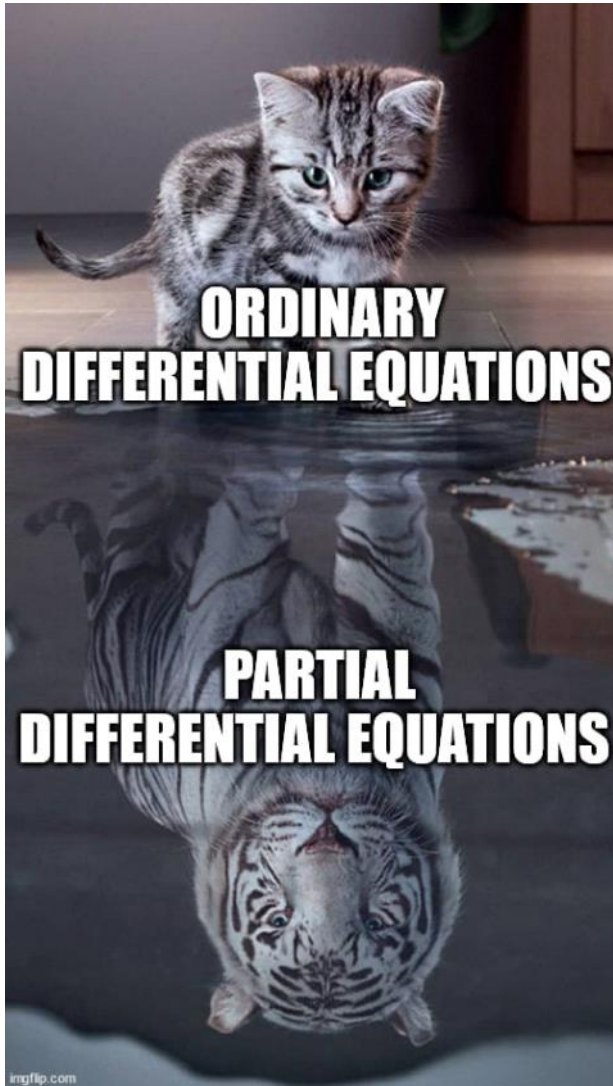
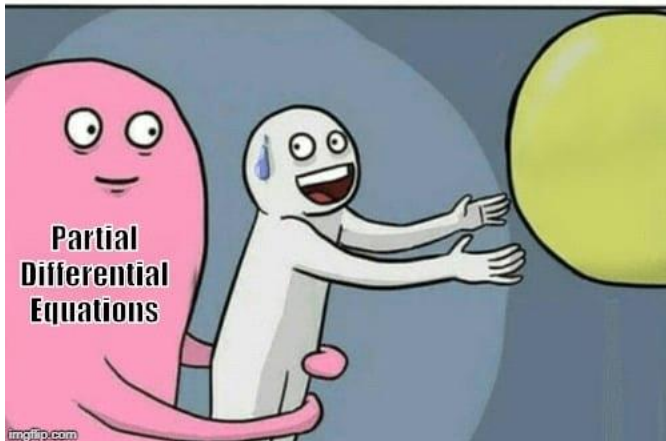
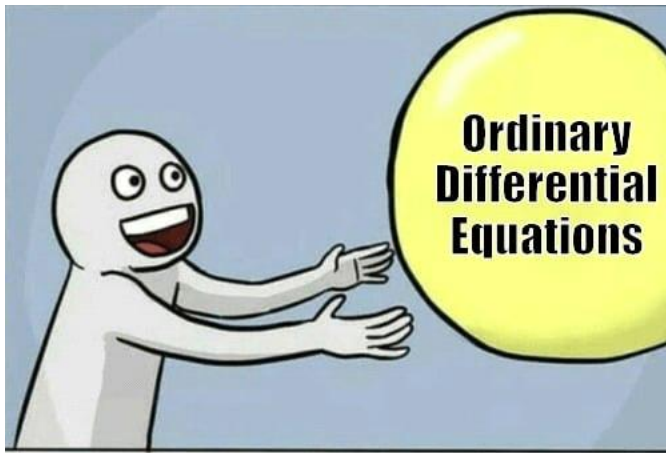


Motivation-2

Saturday, 12 September 2026 4:14 pm

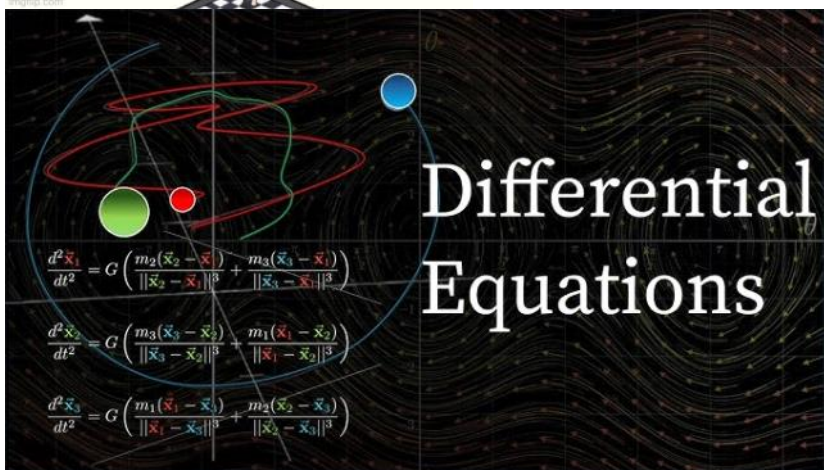
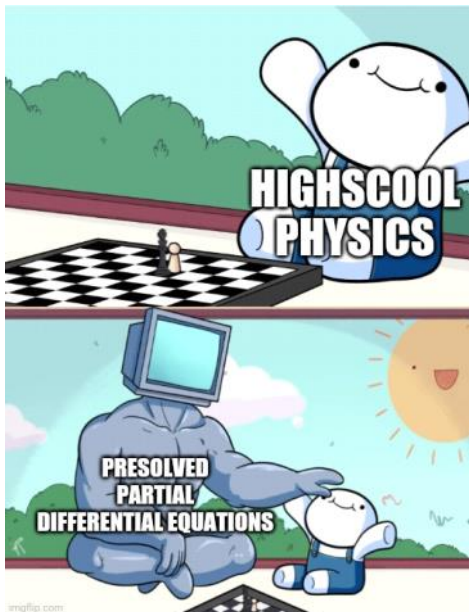




Partially solving a differential equation

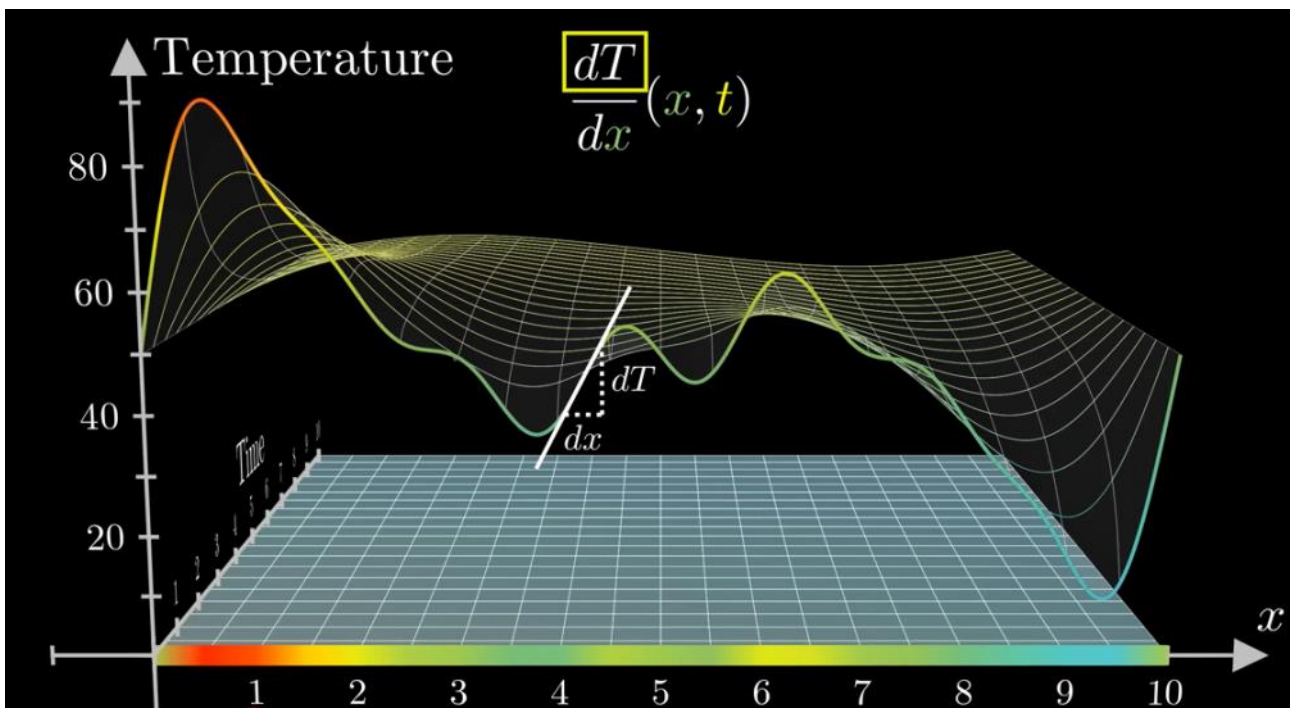
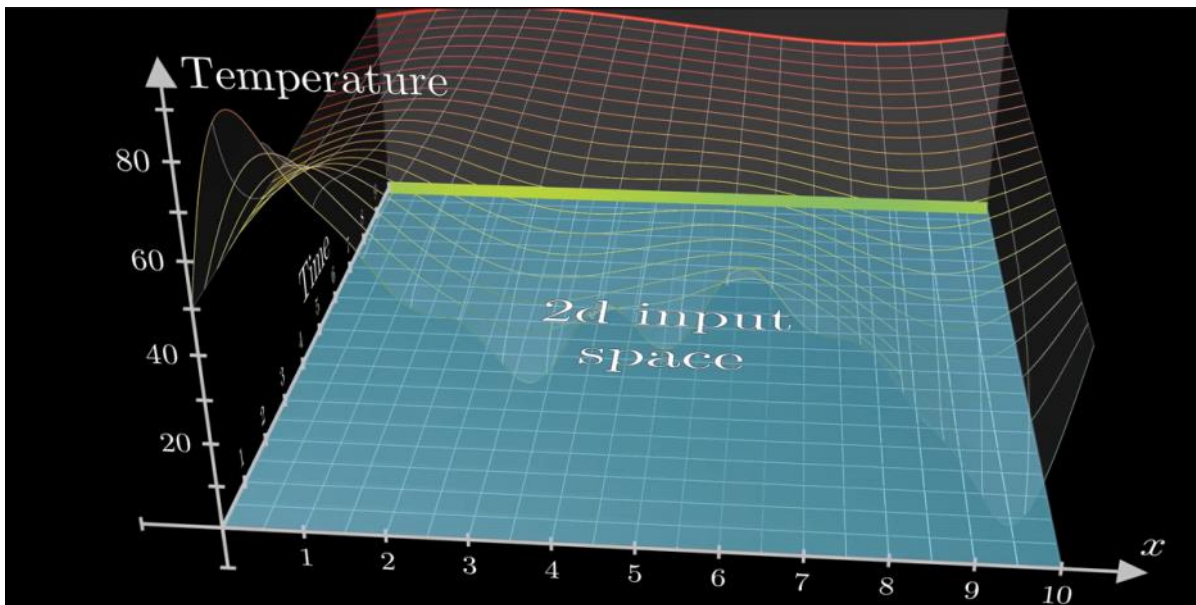
Solving a partial differential equation

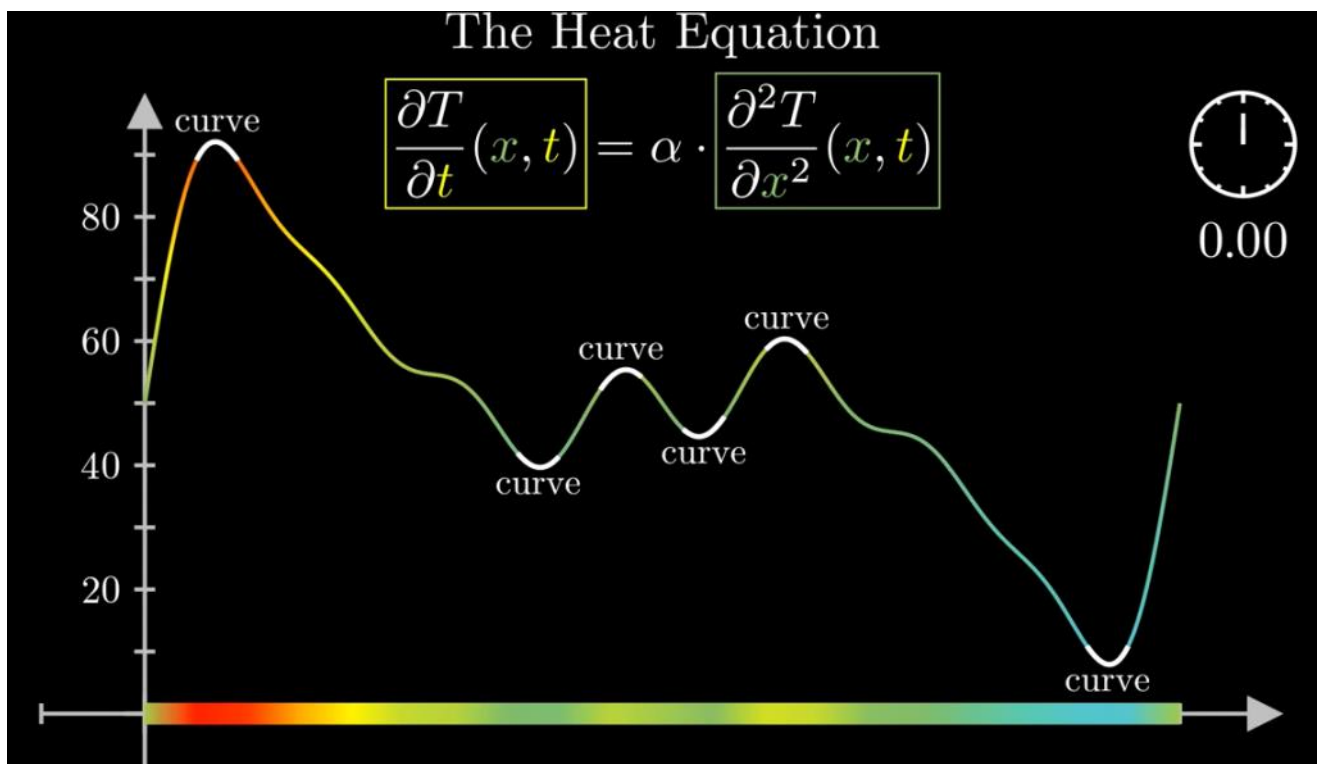
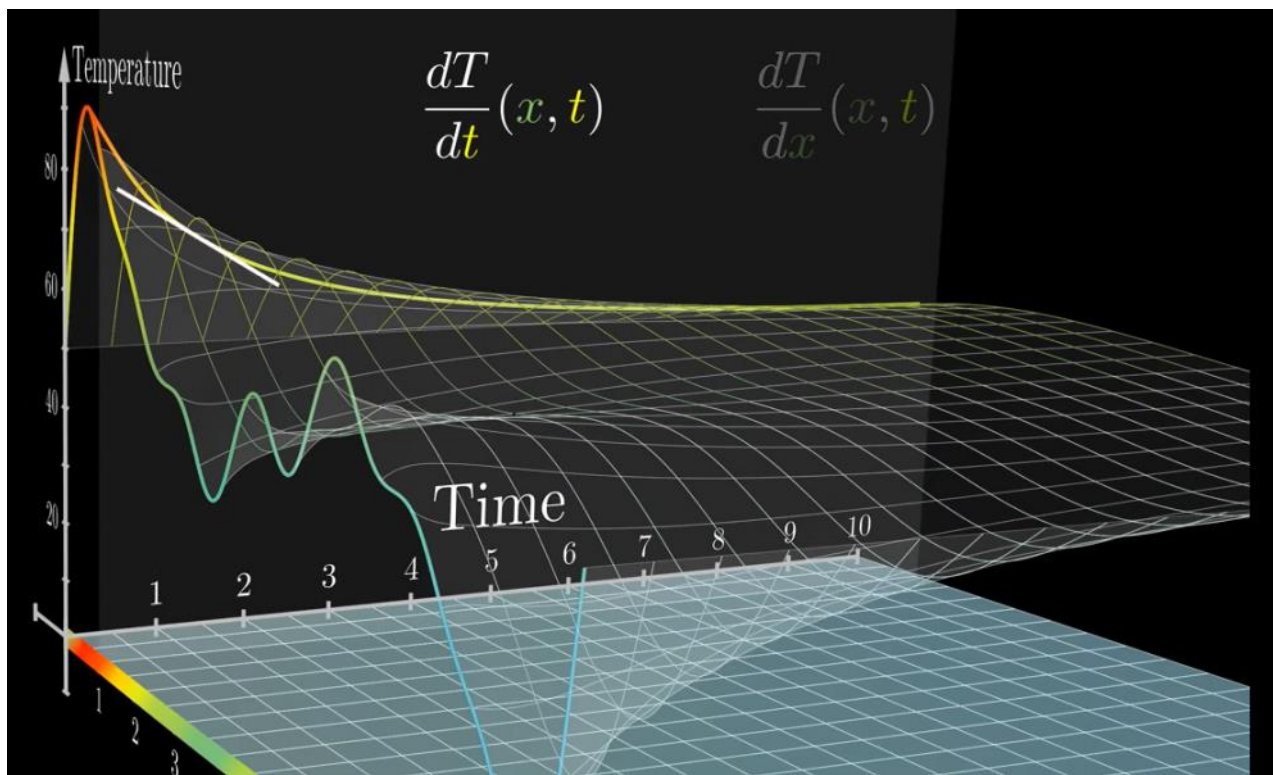




$$\frac{\partial T}{\partial t}(x, t) = \alpha \cdot \frac{\partial^2 T}{\partial x^2}(x, t)$$





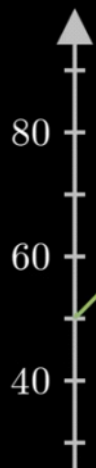


Partial Differential Equation

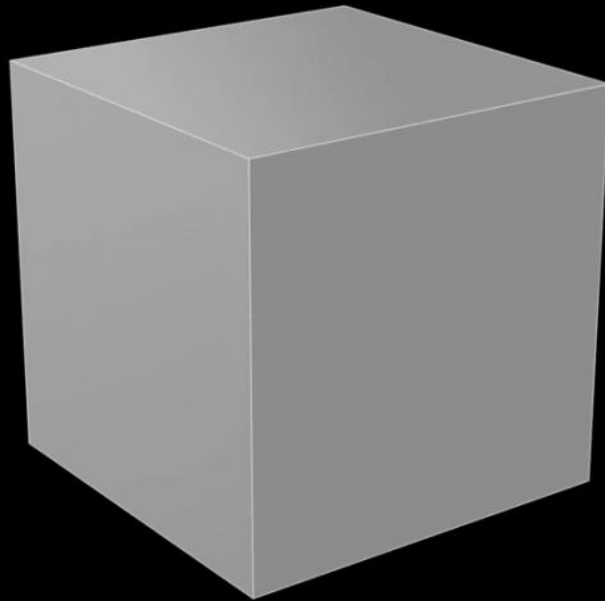
$$\frac{\partial T}{\partial t}(x, t) = \alpha \cdot \frac{\partial^2 T}{\partial x^2}(x, t)$$

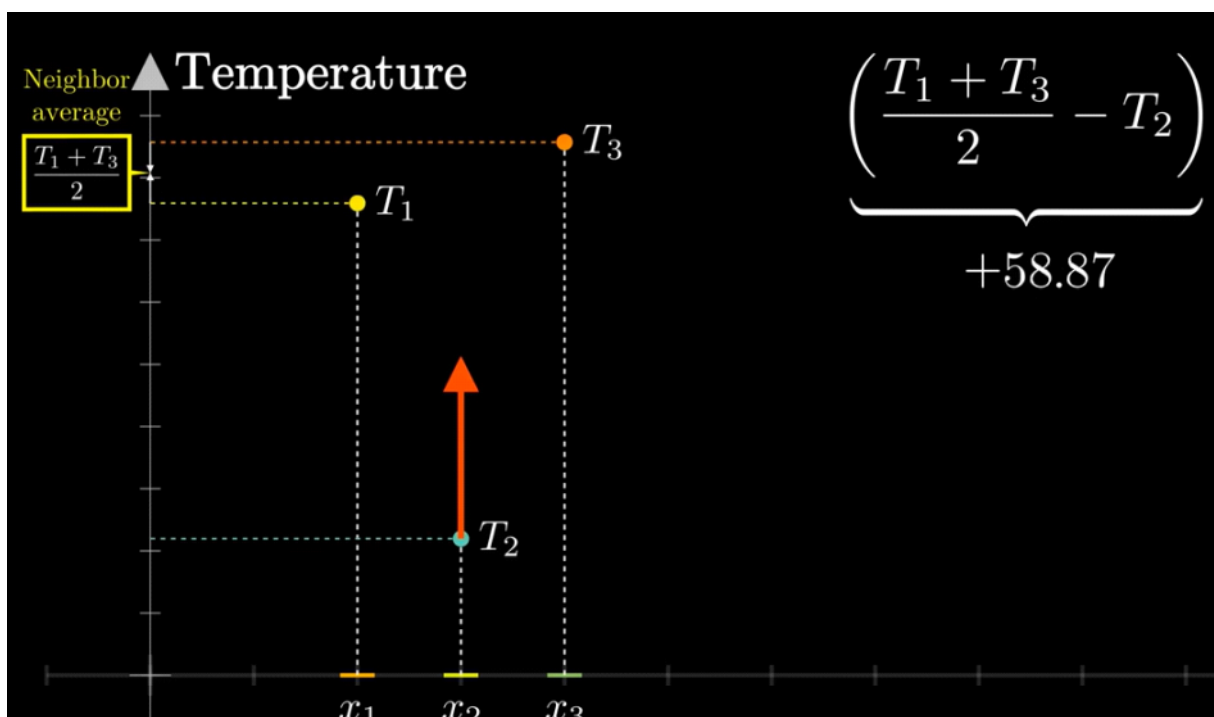
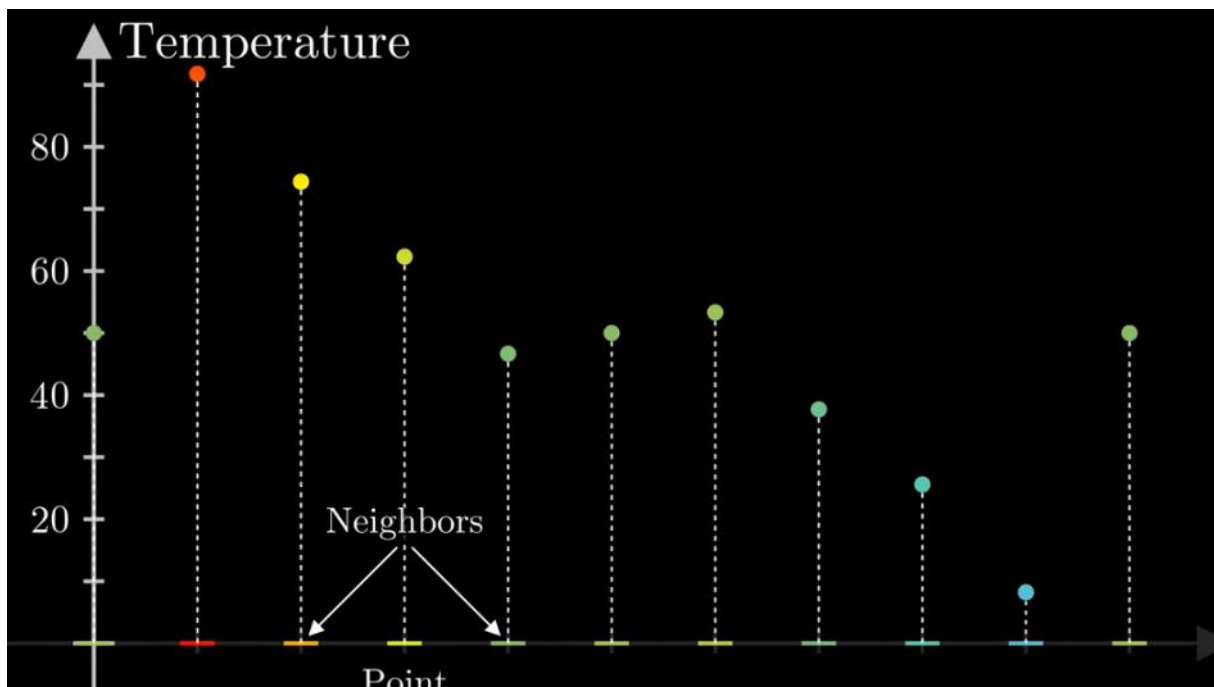


6.83



$$\frac{\partial T}{\partial t}(x, y, z, t) = \alpha \left(\frac{\partial^2 T}{\partial x^2}(x, y, z, t) + \frac{\partial^2 T}{\partial y^2}(x, y, z, t) + \frac{\partial^2 T}{\partial z^2}(x, y, z, t) \right)$$



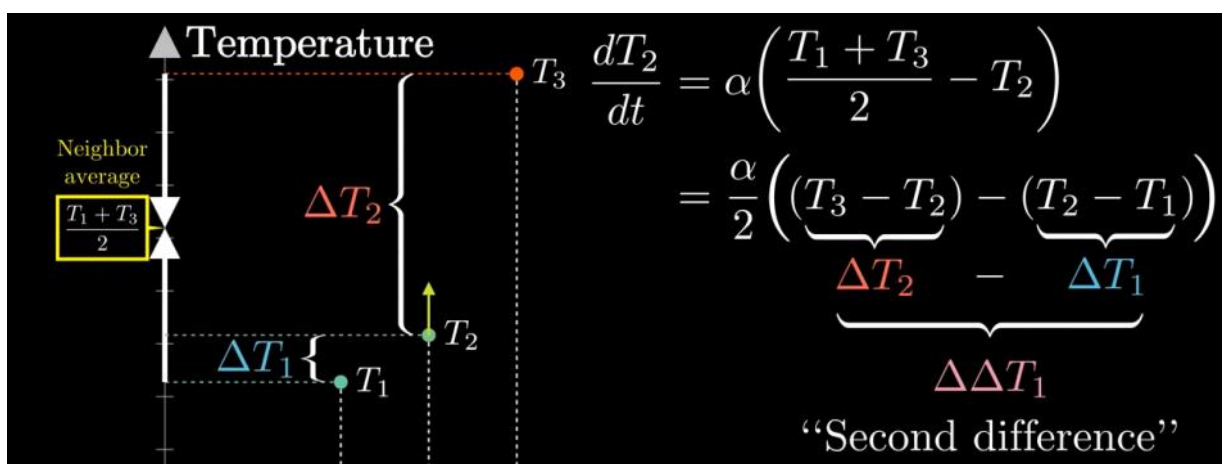
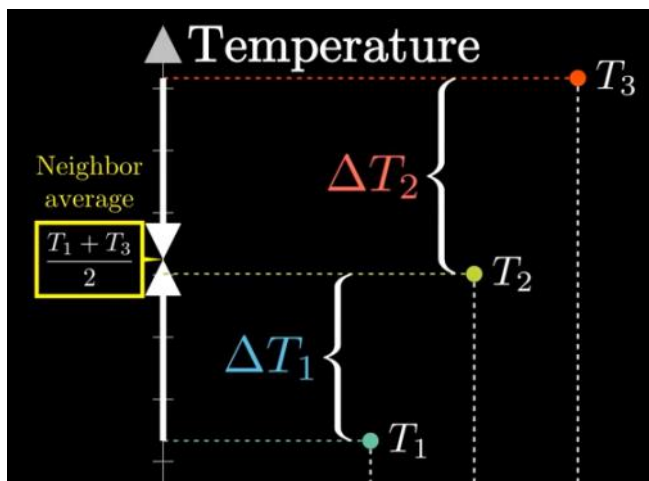
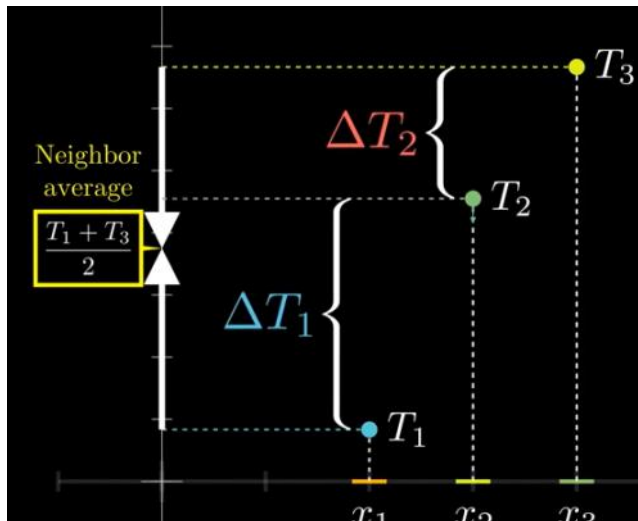


$$\frac{dT_2}{dt} = \alpha \left(\frac{T_1 + T_3}{2} - T_2 \right)$$

$$= \frac{\alpha}{2} \left((T_3 - T_2) - (T_2 - T_1) \right)$$

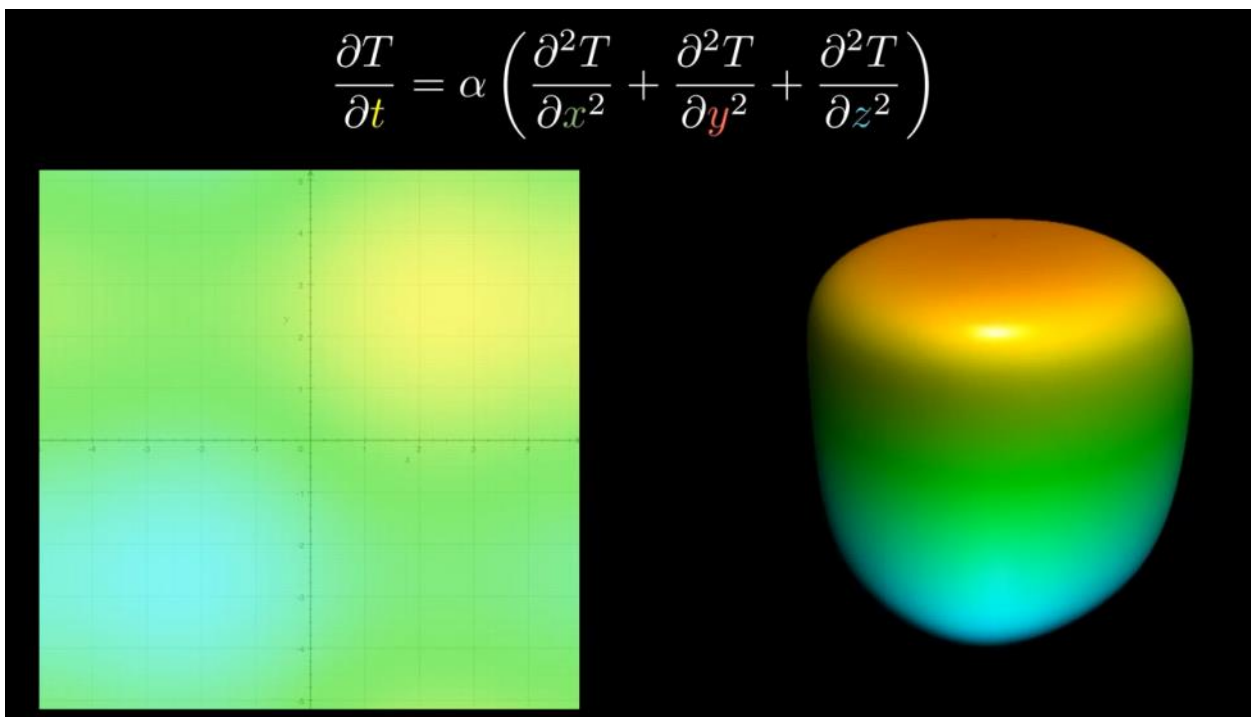
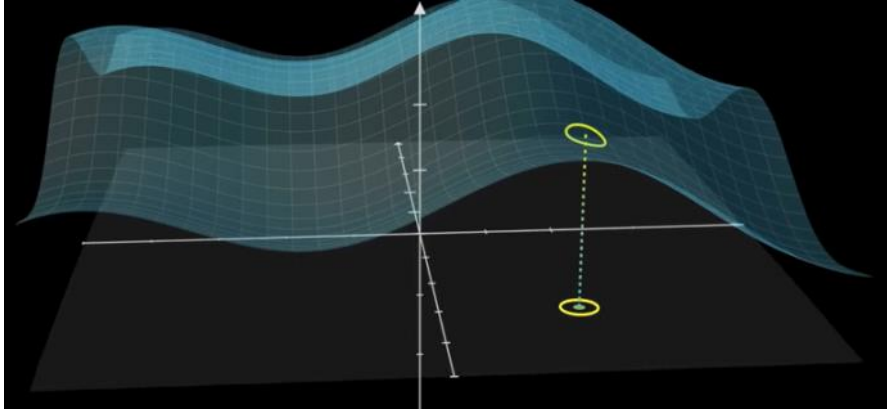
$$= \frac{\alpha}{2} \left(\underbrace{(T_3 - T_2)}_{\Delta T_2} - \underbrace{(T_2 - T_1)}_{\Delta T_1} \right)$$

Difference of differences



$$\frac{\partial T}{\partial t} = \alpha \underbrace{\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right)}_{\nabla^2 T}$$

“Laplacian”



Partial differential equations

In all the above examples we had a function y of a single variable x, y being the solution of an **ordinary** differential equation.

In engineering and science ODEs arise as models for systems where there is **one** independent variable (often x) and **one** dependent variable (often y). Obvious examples are lumped electrical circuits where the current i is a function only of time t (and not of position in the circuit) and lumped mechanical systems (such as the simple harmonic oscillator referred to above) where the displacement of a moving particle depends only on t .

Classification of PDEs

$$\boxed{A} \frac{\partial^2 u}{\partial x^2} + \boxed{B} \frac{\partial^2 u}{\partial x \partial y} + \boxed{C} \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu + G = 0$$

$$B^2 - 4AC < 0 \quad \text{Elliptic PDE}$$

$$B^2 - 4AC = 0 \quad \text{Parabolic PDE}$$

$$B^2 - 4AC > 0 \quad \text{Hyperbolic PDE}$$

Three fundamental examples

Wave equation (hyperbolic)

Laplace equation (elliptic)

Heat equation (parabolic)

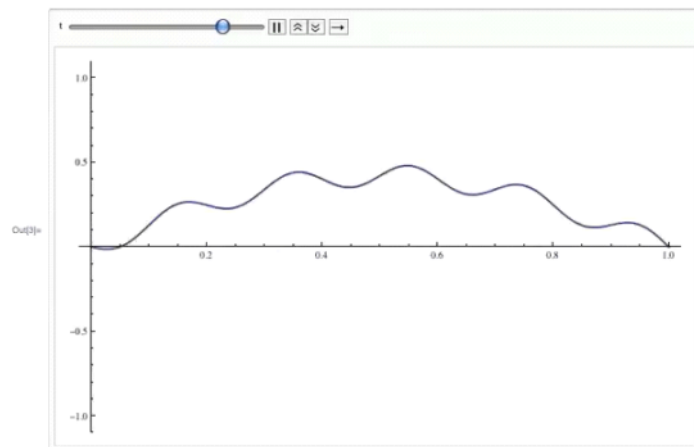
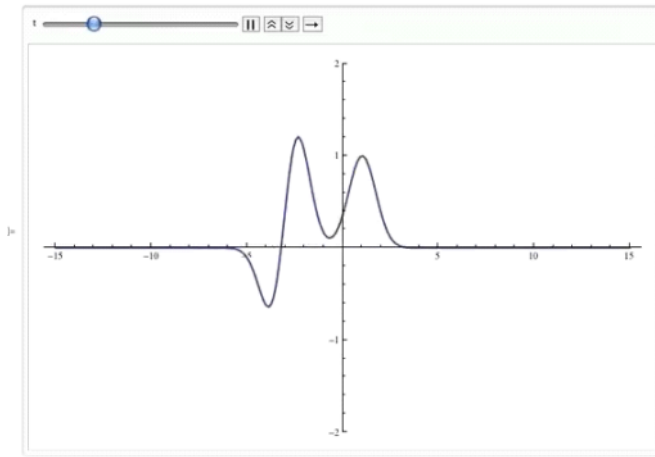
Wave equation

$$u_{tt} = u_{xx}$$

d'Alembert 1740's

water wave

vibrating string



Laplace equation

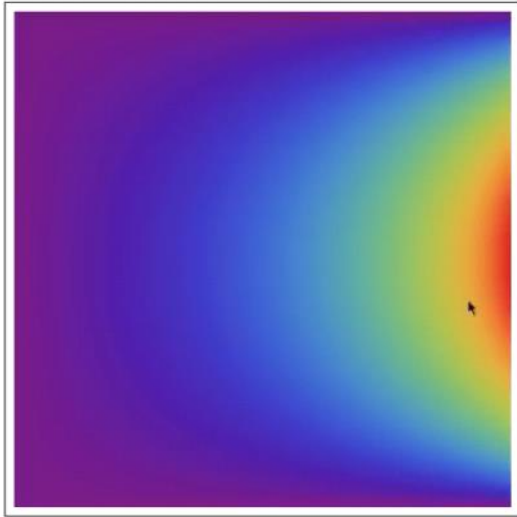
$$u_{xx} + u_{yy} = 0$$

Laplace 1780's

mechanical equilibrium



thermal equilibrium



6) Parabolic, Hyperbolic, or Elliptic: (for linear, 2nd order, PDEs)

$$A \frac{\partial^2 f}{\partial x^2} + B \frac{\partial^2 f}{\partial x \partial y} + C \frac{\partial^2 f}{\partial y^2} + D \frac{\partial f}{\partial x} + E \frac{\partial f}{\partial y} + F f = G \quad \text{Equation (A)}$$

Hyperbolic: $B^2 - 4AC > 0$: variable substitutions to obtain:

$$\frac{\partial^2 f}{\partial \eta^2} - \frac{\partial^2 f}{\partial \xi^2} + g = 0, \quad g = \text{first order / lower terms}$$

$$x^2 - y^2 = 1 \quad (\text{hyperbola})$$

Parabolic: $B^2 - 4AC = 0$: variable substitutions:

$$\frac{\partial^2 f}{\partial \eta^2} + g = 0 \quad \Leftrightarrow \quad x^2 + y = 0$$

Elliptic: $B^2 - 4AC < 0$, variable substitutions:

$$\frac{\partial^2 f}{\partial \eta^2} + \frac{\partial^2 f}{\partial \xi^2} + g = 0 \quad \Leftrightarrow \quad x^2 + y^2 = 1 \quad (\text{ellipse})$$

Example 1

Show that $u = \sin x \cosh y$ satisfies the PDE $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

This PDE is known as **Laplace's equation in two dimensions** and it arises in many applications e.g. electrostatics, fluid flow, heat conduction.

Solution

$$u = \sin x \cosh y \quad \Rightarrow \quad \frac{\partial u}{\partial x} = \cos x \cosh y \quad \text{and} \quad \frac{\partial u}{\partial y} = \sin x \sinh y$$

$$\text{Differentiating again gives } \frac{\partial^2 u}{\partial x^2} = -\sin x \cosh y \quad \text{and} \quad \frac{\partial^2 u}{\partial y^2} = \sin x \cosh y$$

Hence

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = -\sin x \cosh y + \sin x \cosh y = 0$$

so the given function $u(x, y)$ is indeed a solution of the PDE.

$$\text{Show that } u = e^{-2\pi^2 t} \sin \pi x \text{ is a solution of the PDE } \frac{\partial^2 u}{\partial x^2} = \frac{1}{2} \frac{\partial u}{\partial t}$$

Answer

$$\frac{\partial u}{\partial t} = -2\pi^2 e^{-2\pi^2 t} \sin \pi x \quad \frac{\partial u}{\partial x} = \pi e^{-2\pi^2 t} \cos \pi x$$

Now find $\frac{\partial^2 u}{\partial x^2}$ and complete the Task:

Your solution

Answer

$$\frac{\partial^2 u}{\partial x^2} = -\pi^2 e^{-2\pi^2 t} \sin \pi x, \text{ and we see that } \frac{\partial^2 u}{\partial x^2} = \frac{1}{2} \frac{\partial u}{\partial t} \text{ as required.}$$

The PDE in the above Task has the general form

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{k} \frac{\partial u}{\partial t}$$

where k is a positive constant. This equation is referred to as the **one-dimensional heat conduction equation** (or sometimes as the **diffusion equation**). In a heat conduction context the dependent variable u represents the temperature $u(x, t)$.

The third important PDE involving two independent variables is known as the **one-dimensional wave equation**. This has the general form

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

(Note that both partial derivatives in the wave equation are second-order in contrast to the heat conduction equation where the time derivative is first order.)

Example 2

- (a) Verify that $u(x, t) = u_0 \sin\left(\frac{\pi x}{\ell}\right) \cos\left(\frac{\pi ct}{\ell}\right)$ (where u_0 , ℓ and c are constants) satisfies the one-dimensional wave equation.
- (b) Verify the boundary conditions i.e. $u(0, t) = u(\ell, t) = 0$.
- (c) Verify the initial conditions i.e. $\frac{\partial u}{\partial t}(x, 0) = 0$ and $u(x, 0) = u_0 \sin\left(\frac{\pi x}{\ell}\right)$.
- (d) Give a physical interpretation of this problem.

(a) By straightforward partial differentiation of the given function $u(x, t)$:

$$\begin{aligned} \frac{\partial u}{\partial x} &= u_0 \frac{\pi}{\ell} \cos\left(\frac{\pi x}{\ell}\right) \cos\left(\frac{\pi ct}{\ell}\right) & \frac{\partial^2 u}{\partial x^2} &= -u_0 \left(\frac{\pi}{\ell}\right)^2 \sin\left(\frac{\pi x}{\ell}\right) \cos\left(\frac{\pi ct}{\ell}\right) \\ \frac{\partial u}{\partial t} &= -u_0 \left(\frac{\pi c}{\ell}\right) \sin\left(\frac{\pi x}{\ell}\right) \sin\left(\frac{\pi ct}{\ell}\right) & \frac{\partial^2 u}{\partial t^2} &= -u_0 \left(\frac{\pi c}{\ell}\right)^2 \sin\left(\frac{\pi x}{\ell}\right) \cos\left(\frac{\pi ct}{\ell}\right) \end{aligned}$$

We see that $\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$ which completes the verification.

(b) Putting $x = 0$, and leaving t arbitrary, in the given solution for $u(x, t)$ gives

$$u(x, 0) = u_0 \sin 0 \cos\left(\frac{\pi ct}{\ell}\right) = 0 \quad \text{for all } t$$

Similarly putting $x = \ell$, t arbitrary: $u(\ell, 0) = u_0 \sin \pi \cos\left(\frac{\pi ct}{\ell}\right) = 0 \quad \text{for all } t$

(c) Evaluating $\frac{\partial u}{\partial t}$ firstly for general x and t

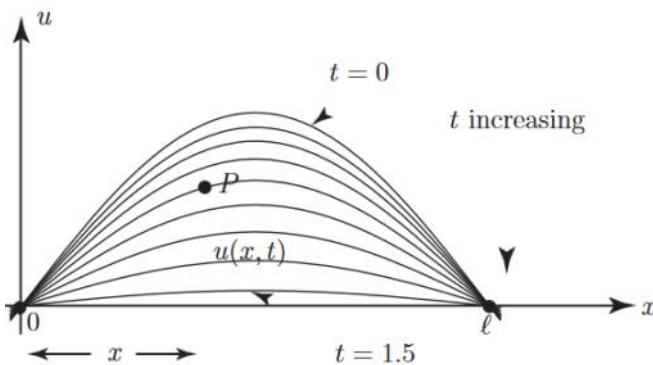
$$\frac{\partial u}{\partial t} = -u_0 \left(\frac{\pi c}{\ell}\right) \sin\left(\frac{\pi x}{\ell}\right) \sin\left(\frac{\pi ct}{\ell}\right)$$

Now putting $t = 0$ leaving x arbitrary

$$\frac{\partial u}{\partial t}(x, 0) = -u_0 \left(\frac{\pi c}{\ell}\right) \sin\left(\frac{\pi x}{\ell}\right) \sin 0 = 0.$$

Also, putting $t = 0$ in the expression for $u(x, t)$ gives

$$u(x, 0) = u_0 \sin\left(\frac{\pi x}{\ell}\right) \cos 0 = u_0 \sin\left(\frac{\pi x}{\ell}\right).$$





Key Point 4

The three PDEs of greatest general interest involving two independent variables are

- (a) The two-dimensional Laplace equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

- (b) The one-dimensional heat conduction equation:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{k} \frac{\partial u}{\partial t}$$

- (c) The one-dimensional wave equation:

$$\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

SOLUTION OF EQUATION BY DIRECT INTEGRATION

Example 3. Solve $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos (2x + 3y)$

Solution. $\frac{\partial^3 z}{\partial x^2 \partial y} = \cos (2x + 3y)$

Integrating w.r.t. 'x', we get

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{1}{2} \sin (2x + 3y) + f(y)$$

Integrating w.r.t. x, we get

$$\frac{\partial z}{\partial y} = -\frac{1}{4} \cos (2x + 3y) + x \int f(y) dy + g(y)$$

$$= -\frac{1}{4} \cos (2x + 3y) + x \phi(y) + g(y)$$

Integrating w.r.t. 'y', we get

$$z = -\frac{1}{12} \sin (2x + 3y) + x \int \phi(y) dy + \int g(y) dy$$

$$z = -\frac{1}{12} \sin (2x + 3y) + x \phi_1(y) + \phi_2(y) \quad \text{Ans.}$$

Example 4. Solve $\frac{\partial^2 z}{\partial x \partial y} = x^2 y$

subject to the condition $z(x, 0) = x^2$ and $z(1, y) = \cos y$.

Solution. $\frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = x^2 y$

$$z = \frac{x^3 y^2}{6} + F(y) + g(x)$$

Condition 1: Putting $z = x^2$ and $y = 0$ in (1), we get

$$x^2 = 0 + F(0) + g(x)$$

Putting the value of $g(x)$ in (1), we get

$$z = \frac{x^3 y^2}{6} + F(y) + x^2 - F(0)$$

Condition 2: $z(1, y) = \cos y$

Putting $x = 1$ and $z = \cos y$ in (2), we get

$$\cos y = \frac{y^2}{6} + F(y) + 1 - F(0)$$

Putting the value of $F(y)$ in (2), we obtain

$$z = \frac{1}{6} x^3 y^2 + \cos y - \frac{1}{6} y^2 - 1 + F(0) + x^2 - F(0)$$

$$z = \frac{1}{6} x^3 y^2 + \cos y - \frac{1}{6} y^2 - 1 + x^2 \quad \text{Ans.}$$

Example 5. Solve $\frac{\partial^2 z}{\partial y^2} = z$, if $y = 0$, $z = e^x$ and $\frac{\partial z}{\partial y} = e^{-x}$

Solution. If z is a function of y alone, then

$$z = A \sinh y + B \cosh y \text{ where } A \text{ and } B$$

and constants, since z is a function of x and y , A and B are the functions of x .

Then $z = \sinh y \cdot f(x) + \cosh y \cdot \phi(x)$

On putting $y = 0$ and $z = e^x$ in (1), we obtain

$$e^x = \phi(x)$$

(1) becomes $z = \sinh y \cdot f(x) + \cosh y \cdot e^x$

On differentiating (2) w.r.t. y , we get

$$\frac{\partial z}{\partial y} = \cosh y \cdot f(x) + \sinh y \cdot e^x$$

On putting $y = 0$ and $\frac{\partial z}{\partial y} = e^{-x}$ in (3), we obtain

$$e^{-x} = f(x)$$

(2) becomes,

$$z = e^{-x} \sinh y + e^x \cosh y \quad \mathbf{Ans.}$$